

Dislocated Bilayer MOF Enables High-Selectivity Photocatalytic Reduction of CO₂ to CO

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The highly selective photoreduction of CO₂ into valuable small-molecule chemical feedstocks such as CO is an effective strategy for addressing the energy crisis and environmental problems. However, it remains a challenge because the complex CO₂ photoreduction process usually generates multiple possible products and requires a subsequent separation step. In this paper, 2D monolayer and bilayer porphyrin-based metal-organic frameworks (MOFs) are successfully constructed by adjusting the reaction temperature and solvent polarity with 5,10,15,20-tetrakis(4-pyridyl)porphyrin as the light-harvesting ligand. The bilayer MOF is a low-dimensional MOF with a special structure in which the upper and lower layers are arranged in dislocation and are bridged by halogen ions. This bilayer MOF exhibits 100% ultra-high selectivity for the reduction of CO₂ to CO under simulated sunlight without any cocatalyst or photosensitizer and can be recycled at least three times. The intrinsic mechanism of this photocatalytic CO₂ reduction process is explored through experimental characterization and density functional theory (DFT) calculations. This work shows that the rational design of the number of layers in 2D MOF structures can tune the stability of these structures and opens a new avenue for the design of highly selective MOF photocatalysts.

solar energy into storable high-density chemical energy (such as solar fuel) and realize CO₂ resource utilization, thereby simultaneously solving the problems of energy crisis and global temperature warming, which is an emerging and reliable CO₂ emission reduction technology. However, the current traditional photocatalysts for CO₂ reduction are limited by the weak light absorption ability, difficult adsorption and activation of CO₂, serious electron-hole recombination, and poor stability, etc., so it is difficult to achieve a satisfactory reduction effect.^[5,14–18] The highly active CO₂ adsorption and activation sites and good photosensitive structures are two characteristics that need to be optimized in the design and synthesis of photocatalysts.^[16,19]

Metal-organic frameworks (MOFs) are porous crystalline materials with diverse properties such as tunable chemical compositions and crystal structures, highly ordered light collectors, catalytic sites,

high porosity, and desirable CO₂ affinity, and are widely used as photocatalytic materials.^[20–26] In especial, 2D MOFs tend to have higher performance in photocatalysis, due to large surface areas and easily accessible active sites. These properties can reduce the diffusion barrier, facilitate contact between the substrate and active sites, and enable rapid mass transfer and charge transfer.^[27–35] However, there are relatively few examples of pure MOF photocatalysts for CO₂ reduction due to their disadvantages of weak visible light responsiveness and severe photogenerated electron-hole recombination, which limit the application of MOFs in photocatalysis. Strong absorption in visible light region, accessibility of catalytic sites and efficient separation of photoinduced electron-hole pairs are three key factors that should be considered in the construction of MOFs photocatalysts.^[36–40] As one of the most important photosensitive organic compounds, porphyrins have been widely used in the construction of MOF photocatalysts.^[41–50] Porphyrin-integrated MOFs exhibit the inherent light-harvesting properties of porphyrin ligands, typically showing broad absorption in the 200–800 nm range. The introduction of coordinatively unsaturated metal sites into metalloporphyrin-free ligands provides atomically dispersed catalytic centers, significantly facilitates the directional transfer of energy, and improves electron-hole separation efficiency.^[20,31,51–55] Although some MOF-based CO₂ photoreduction catalysts have been reported in the literature so far, their selectivity for the photocatalytic CO₂ reduction reaction

1. Introduction

Over the past decades, the excessive emission of carbon dioxide (CO₂) has led to increasing environmental problems due to the massive consumption of fossil fuels. The efficient use of CO₂ has become a global research hotspot.^[1–5] Compared with energy-intensive CO₂ capture and storage technologies, converting CO₂ into valuable small-molecule chemical feedstocks or energy fuels (such as CO, CH₄, HCOOH, and other chemicals) is an effective strategy for solving both the energy crisis and environmental problems.^[6–13] In particular, CO as a C₁ chemical is an important raw material for the synthesis of various chemical products. Alcohols, ethers, acids and esters and their derivatives can be efficiently prepared from carbon monoxide. Among the many CO₂ conversion technologies, photocatalytic CO₂ reduction technology can efficiently convert

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is still unsatisfactory. Therefore, specific carbon products cannot be obtained with these catalysts, leading to the potential production of a variety of products such as CO, CH₄, HCOOH, and CH₃OH.^[20,31,53,56] The formation of these products is also usually accompanied by the hydrogen evolution reaction. Furthermore, in the absence of cocatalysts or photosensitizers, the efficiency of MOFs for the photocatalysis of CO₂ is relatively low.^[12,57,58] Consequently, designing and constructing mild catalyst systems with high selectivity and efficiency without the participation of additional cocatalysts or photosensitizers is crucial for developing efficient CO₂ reduction systems.

In general, the properties and functions of 2D MOFs are highly dependent on the composition and structure of their 2D layers as well as the controllable arrangement and thickness or number of 2D layers. Different numbers of atomic layers can lead to different structures and electronic/optical properties due to stacking effects and structural stability. These factors can greatly alter the pathway and performance of catalysts.^[54,59–63] Therefore, in this work, a novel strategy for the construction of 2D monolayer and bilayer porphyrin-based MOFs is proposed. Both the monolayer and bilayer MOFs significantly promote the photoreduction of CO₂ to CO without any cocatalyst or photosensitizer. It is worth noting that the bilayer MOF exhibits good structural stability and can reduce CO₂ to CO with 100% selectivity, outperforming most other reported MOF photocatalysts. Furthermore, the intrinsic mechanism of this photocatalytic CO₂ reduction process was explored by experimental characterization and DFT calculations. This work opens new avenues for constructing 2D bilayer MOF to achieve efficient photocatalytic materials with high selectivity and high stability.

2. Result and Discussion

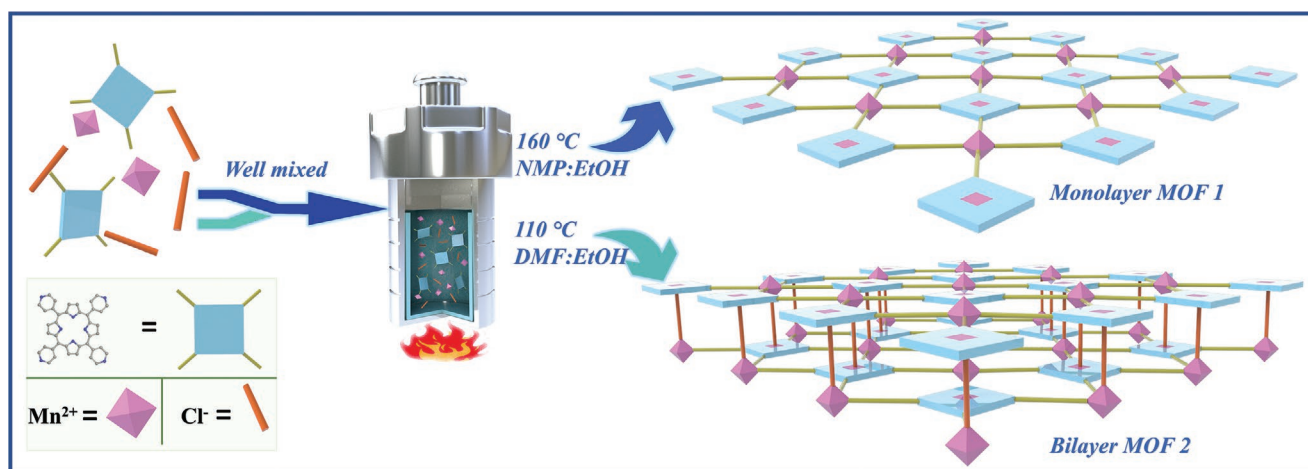
Both the bilayer and monolayer MOFs are freestanding 2D porphyrin-based materials by assembling TPP monomers with manganese metal ions. As shown in **Scheme 1**, when DMF/EtOH solvent was as the reaction solvent and the solvothermal temperature was adjusted to 110 °C, high-quality large-area 2D bilayer MOF 2 was directly obtained with a high yield (48%,

based on TPP). Monolayer porphyrin-based MOF 1 can be obtained at 160 °C when NMP/EtOH was used as the reaction solvent (Yield, 10%, based on TPP). But, due to the thin layered structure of these monolayer MOF samples, it was difficult to obtain crystal samples with high crystalline quality. The obtained bulk MOF materials were used as precursors, and the corresponding ultrathin MOF nanosheets (nMOF) were obtained by ultrasonic exfoliation.

2.1. Descriptions of Crystal Structures

2.1.1. Crystal Structure of Bilayer MOF 2

The single-crystal structures of MOF 1 and MOF 2 were shown in **Figure 1**. Monolayer MOF 1 is crystallized in the tetragonal system with the *I*₄/*amd* space group, and bilayer MOF 2 is crystallized in monoclinic system with space group *P*₂₁/*m*. Their asymmetric unit contains two types of manganese ions, three chlorides, and one half of a ligand. X-ray structural analysis reveals that MOF 1 exhibits a 2D monolayer structure (Figure 1b,c). Particularly, MOF 2 exhibits a novel 2D bilayer configuration (Figure 1e,f). The manganese center of MOF 1 or MOF 2 is surrounded by two chlorides and four nitrogen atoms, adopting an octahedral configuration. Compared with the N–Mn–N angle in monolayer MOF 1 with 90°, the N–Mn–N angle in MOF 2 slightly deviates with values ranging from 85.133(2)° to 96.337(2)°. In particular, MOF 2 exhibits a low-dimensional MOF structure with a very unique structure, in which, two Mn ions located in the upper and lower layers are further bridged by Cl^{III}, extending the framework into a 2D bilayer framework (Figure 1e). A regular, approximately square grid exists in the 2D dislocated bilayer structure, and the slight deformation of the square grid is related to the tilt of the pyridine ring. As observed in Figure 1f, the thickness of the bilayer is 10.11 Å. The stacking distance between the average planes of the porphyrin macrocycles in adjacent bilayers is about 4.397(8) Å, and the corresponding Mn2–Mn2 separation degree is about 8.5880 Å. Adjacent bilayers are stacked together in an “ABA” mode to facilitate the final 3D stacked structure, leaving



Scheme 1. Schematic diagram of the synthetic routes for preparing monolayer MOF and bilayer MOF.

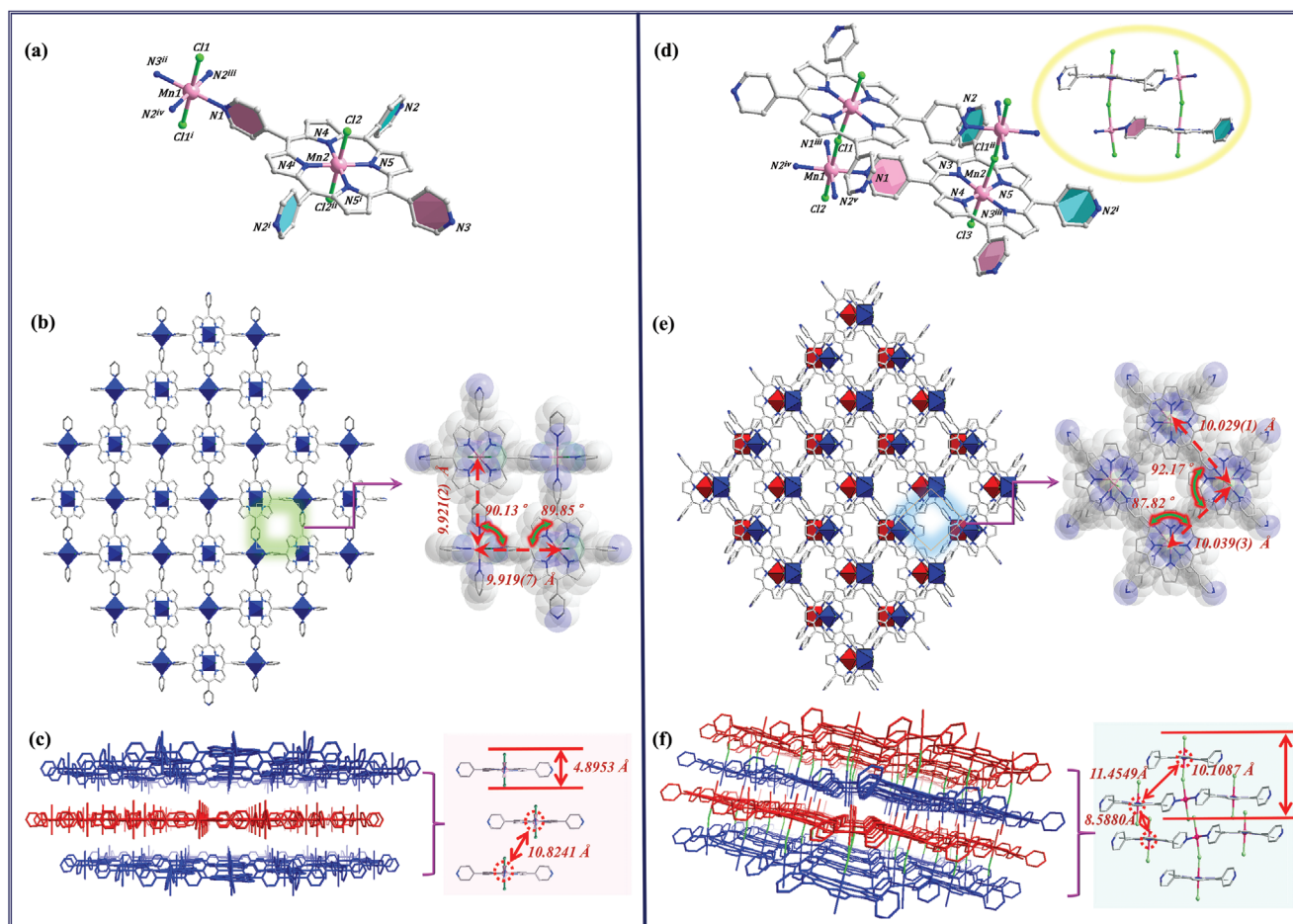


Figure 1. Crystal structures of monolayer MOF 1 and bilayer MOF 2. a) The asymmetric unit of monolayer MOF 1 (Symmetry codes: i: 1-x, y, 1.5-z; ii: x, -1+y, z; iii: 1.5-x, -0.5+y, 1.5-z; iv: -0.5+x, -0.5+y, z; v: x, 1+y, z; vi: 0.5+x, 0.5+y, z). b) View of the single layer of MOF 1 along c direction (illustration: square grid in structure). c) Side view of the stacking 2D layers along b direction. d) The asymmetric unit of bilayer MOF 2 (Symmetry codes: i: x, 1.5-y, z; ii: 1-x, 1-y, 1-z; iii: x, 0.5-y, z; iv: -1+x, 0.5-y, -1+z; v: -1+x, y, -1+z; vi: 1+x, y, 1+z). e) View of single layer of MOF 2 along a direction (illustration: square grid in structure). f) Side view of the stacking 2D layers along c direction.

channels between adjacent layers. According to PLATON calculation results, the accessible volume of the solvent in MOF 2 is estimated to be 22.5% (443.3 Å³ per unit cell).

In order to understand the structural characteristics of bilayer MOF 2 more intuitively, topology analysis is used to simplify each node. Unlike the topological structure of monolayer MOF 1, both TPP and metal ions can be simplified as 5-c node, the overall structure of MOF 2 can be simplified as a novel (5,5)-connected topological network, and the corresponding topological symbol is {4⁸·6²} (Figure S7, Supporting Information). As for monolayer MOF 1 it can be regarded as a 4-connected uninodal network with the point symbol {4⁴·6²} (Figure S4, Supporting Information), similar structural motifs have been seen in previously reported MOFs, such as [(M₂)TpyP] (M = Pd²⁺, Cd²⁺),^[64] [(MnCl₂)TpyP],^[65] and [FeCl₂(L)].^[47] A more detailed crystal structure description of MOF 1 can be found in Supporting Information. It is worth noting that the large conjugation plane of the porphyrin ligand lays the foundation for the construction of the 2D MOF structure. Chloride ions play a vital role as the bridging point between the layers during the construction of the bilayer structure. Therefore, only

linkers that can provide bridging action for the construction of bilayer MOF materials.

2.1.2. Crystal Structure Comparison of Monolayer MOF 1 and Bilayer MOF 2

From the synthetic point of view, monolayer MOF 1 and bilayer MOF 2 are synthesized under the same conditions except for differences in reaction temperature and solvent. The reaction temperature and the polarity of the solvent affect the coordination ability, coordination mode and conformation of the organic ligand, and therefore have a significant impact on the crystal structure of the porous material,^[66–68] while the oxidation state of the metal ion is also affected. These factors impact the final structural morphology of the MOF. The TPP ligands are spontaneously metallized during the construction of the MOF structure, and the axial position of the metal center coordinates two Cl ions to form the MnCl₂-TPP coordination unit. For MOF 1, the MnCl₂-TPP coordination unit adopts a (κ¹)-(κ¹)-(κ¹)-(κ¹)-μ₄ (Mode I, Figure S8a, Supporting Information) mode to connect

Table 1. Detailed structural comparison of monolayer MOF 1 and bilayer MOF 2.

	Monolayer MOF 1	Bilayer MOF 2
MnCl ₂ -TPP coordination mode	Mode I	Mode II
Metal ion coordination geometry	{MnN ₄ Cl ₂ } octahedron	{MnN ₄ Cl ₂ } octahedron
Metal valence	+2, +4	+2, +3
Mn2...Mn2	10.8241 Å	8.5880 Å, 11.4549 Å
Cl ion coordination mode	η^1	$\eta^1, \mu_z, \eta^1: \eta^1$
Dihedral angle	78.463° and 79.542°	56.422° and 66.808°
TPP conformation	centrosymmetric	axisymmetric
2D architecture	(4,4)-c net	(5,5)-c net
Point (Schläfli) symbol	{4 ⁸ .6 ² }	{4 ⁸ .6 ² }

with the adjacent Mn(II), forming a 2D monolayer architecture with the topological symbol of 4,4-c {4⁴.6²}. In MOF 2, the MnCl₂-TPP coordination unit adopts a (κ^1)-(κ^1)-(κ^1)-(κ^1)-(κ^1)- μ_5 (Mode II, Figure S8b, Supporting Information) mode to coordinate with Mn(II) and construct a 5,5-c {4⁸.6²} 2D bilayer topology. A detailed structural comparison of MOFs 1 and 2 is provided in Table 1. Comparing the reaction conditions and the structures of MOFs 1 and 2 shows that the solvent and reaction

temperature have a crucial influence on the final MOF structure. Furthermore, during the metallation of the porphyrins, Mn(II) is oxidized to a higher valence state by oxygen in the air and solvent to obtain Mn(III)-porphyrin or Mn(IV)-porphyrin.^[69,70] In MOF 1, the valence state of the Mn ion at the porphyrin center is +4; but in MOF 2, it is +3. The valence state of Mn in MOFs 1 and 2 was further verified by XPS (Figure 2 and Figure S9, Supporting Information). We speculate that the higher reaction temperature favors the oxidation of metal ions upto higher valence states during the solvothermal process.

2.2. Morphology Analysis

Optical microscopy images show that monolayer MOF 1 exhibits a square flake shape, while bilayer MOF 2 shows a regular block-like cube morphology (Figure S19, Supporting Information). To investigate the microstructure and morphology of these materials, the bulk and nanosheet MOFs were inspected by scanning electron microscopy (SEM), transmission electron microscopy (TEM), and atomic force microscopy (AFM). The SEM images of MOFs 1 and 2 show the presence of crystals with accordion-like layered structures, and the Energy dispersive X-ray spectroscopy (EDX) mapping images show that Mn, C, N, and Cl elements are uniformly distributed over the material, as shown in Figure 3a, b and Figure S20, Supporting Information.

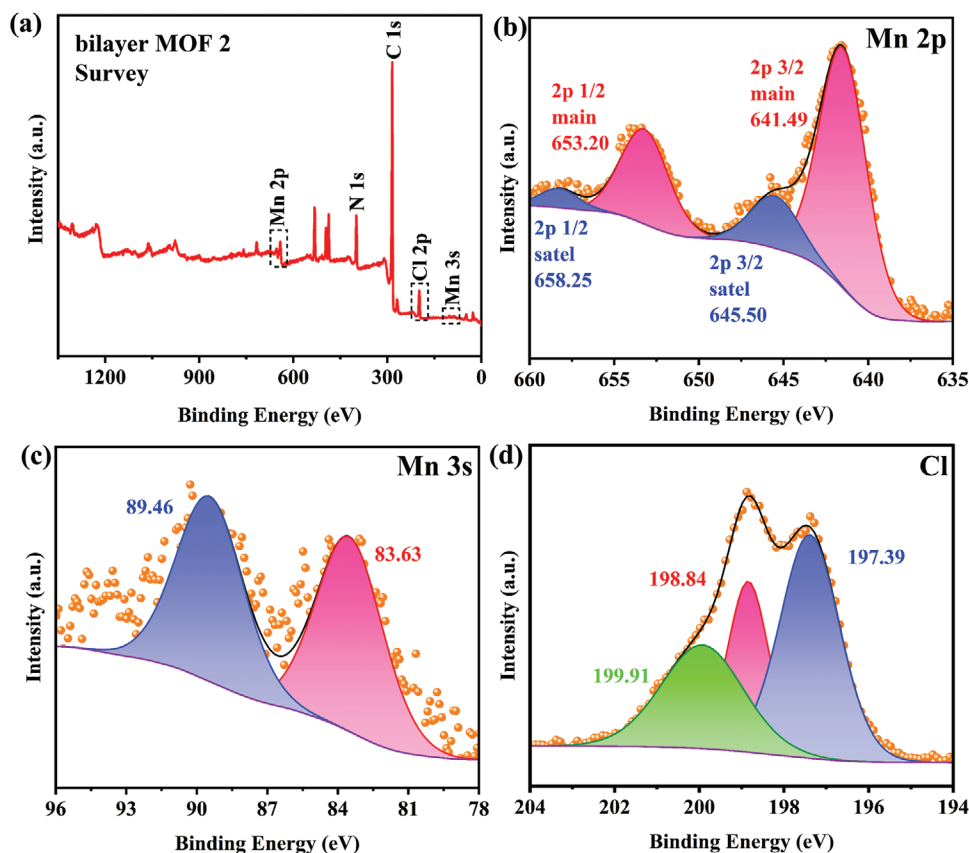


Figure 2. XPS spectra of bilayer MOF 2: a) Survey spectrum of MOF 2, b) Mn 2p spectrum of MOF 2, c) Mn 3s spectrum of MOF 2, d) Cl 2p spectrum of MOF 2.

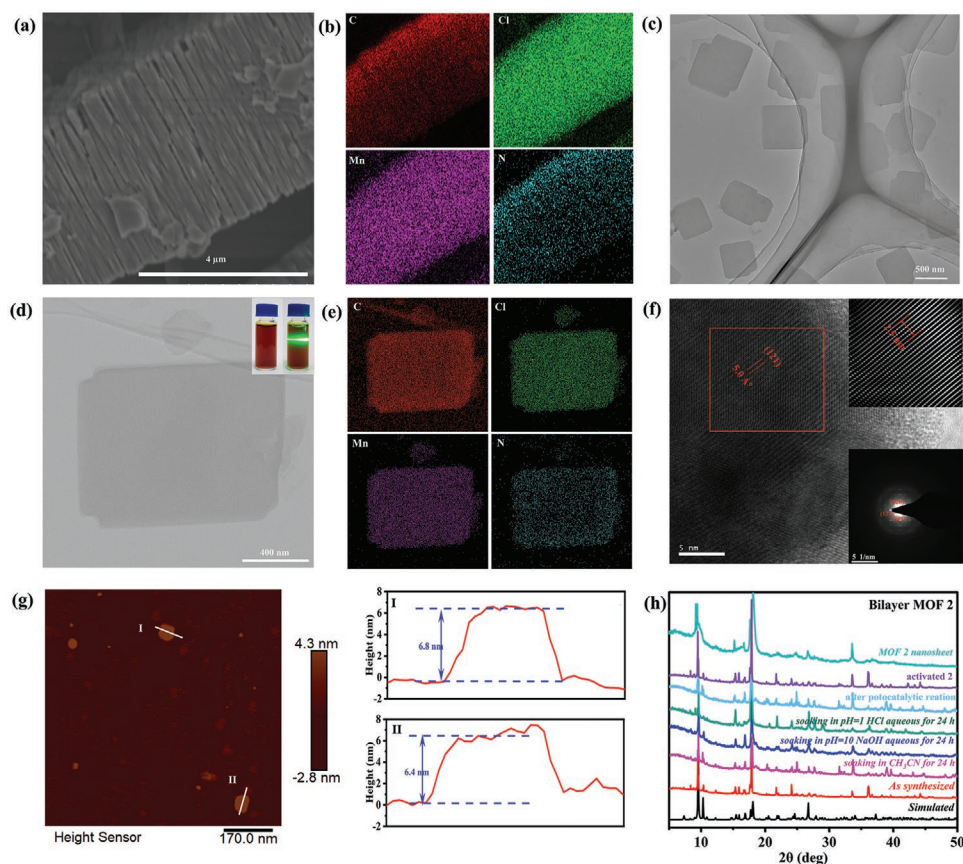


Figure 3. Morphology Characterization and PXRD: a) SEM images of MOF 2, b) SEM-EDS mapping of MOF 2, c,d) TEM image of nMOF 2, e) TEM-EDS mapping of nMOF 2, f) HRTEM of nMOF 2, g) AFM image and corresponding topographies, h) PXRD patterns of MOF 2 and nMOF 2.

Compared with MOF 1, the layer-by-layer stacking of MOF 2 is more compact, which is consistent with the single-crystal diffraction results. TEM was used to confirm the ultrathin structure of the nMOF 1 (Figure S21, Supporting Information) and nMOF 2 (Figure 3c,d). The high-resolution TEM (HRTEM) image of nMOF 2 is shown in Figure 3f. Inverse FFT was performed on the high-resolution image in the red box (inset), which clearly shows the high-resolution lattice stripes of the sample with a grain plane spacing of 5 Å. This fits well with the grain plane in the XRD of the sample (121) and further validates the electron diffraction (SAED) results in the selected region (inset). The thickness of these nMOFs was measured by AFM. As shown in Figure S22, Supporting Information, the thickness of the nMOF 1 is about 2.5 nm. Based on the single-crystal structure of MOF 1, the theoretical interlayer distance is estimated to be 0.49 nm, so the number of layers of the obtained nanosheets is about 5. The thickness of the nMOF 2 is about 6 nm (Figure 3g), corresponding to a thickness of ≈ 6 layers (1.01 nm for a single bilayer). EDX was coupled with the TEM to obtain elemental mapping (Figure 3e and Figure S21, Supporting Information), which confirmed the co-existence of Mn, C, N, and Cl in nMOFs 1 and 2. Moreover, the Mn element was uniformly distributed on the MOF nanosheets. XRD (Figure 3h and Figure S14, Supporting Information) and Fourier transform infrared (FTIR) spectroscopy (Figures S11 and

S12, Supporting Information) demonstrate the consistency of the MOFs bulk and MOF nanosheets.

2.3. Gas Adsorption Studies

The permanent porosity inherent to the MOFs was measured through the reversible adsorption of N_2 at 77 K (Figure 4a). The Brunauer–Emmett–Teller (BET) surface areas of MOFs 1 and 2 are 239.42 and 157.78 $m^2 g^{-1}$, respectively. These values are lower than the theoretically calculated surface areas (372.46 and 214.09 $m^2 g^{-1}$, respectively), which may be attributed to the small amount of free solvent molecules remaining in the pores. Figure 4a shows that both MOFs 1 and 2 exhibit typical type II N_2 adsorption isotherms at 77 K. In the low-pressure region, the sharp increase in N_2 uptake indicates that MOFs 1 and 2 have microporous features. Next, the CO_2 adsorption properties of MOFs 1 and 2 were investigated. At 273 and 298 K, the maximum capacities of MOF 1 for CO_2 are 36.85 and 29.10 $cm^3 g^{-1}$, and the maximum capacities of MOF 2 for CO_2 are 11.86 and 8.03 $cm^3 g^{-1}$, respectively (Figure 4b). The calculated adsorption enthalpies (Q_{st}) of MOFs 1 and 2 are 25.85 and 24.60 $kJ mol^{-1}$, respectively (Figures S17 and S18, Supporting Information). These results indicate the good interaction between the porphyrin MOF framework and CO_2

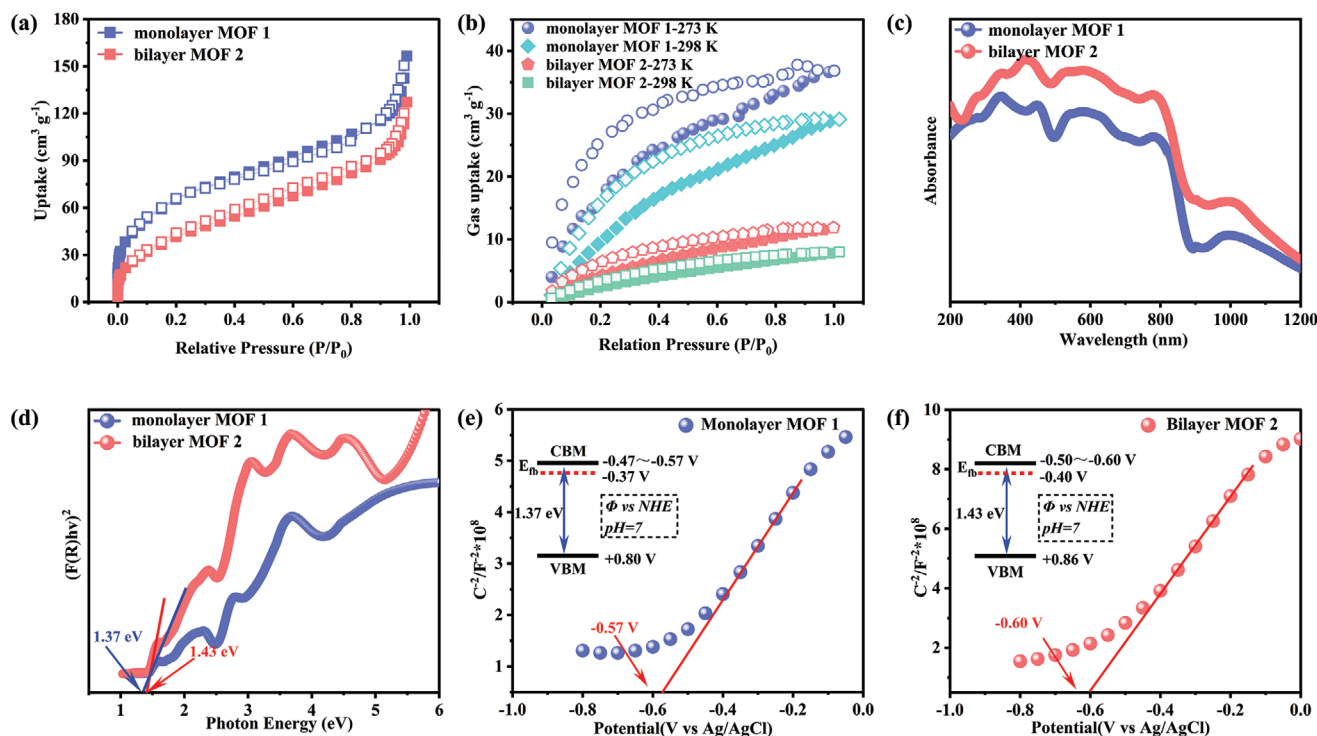


Figure 4. a) N₂ sorption isotherms of MOFs 1 and 2 at 77 K, b) CO₂ sorption isotherms of MOFs 1 and 2, c) Diffuse reflectance spectra of MOFs 1 and 2, d) Diffuse reflectance spectra of MOFs 1 and 2 via K-M transformation, e) Mott-Schottky plot of an electrode containing MOF 1 and calculated redox potentials, f) Mott-Schottky plot of an electrode containing MOF 2 and calculated redox potentials.

molecules, which is beneficial for improving their photocatalytic CO₂ reduction efficiency.

2.4. Thermodynamic Feasibility for CO₂ Reduction

The UV-vis spectra (Figure 4c) of powder samples of MOFs 1 and 2 showed strong absorption properties from 200 to 800 nm, due to the light-harvesting feature of the porphyrinic macrocycle. By fitting the Tauc function to the *K-M* absorbance, the corresponding bandgap energies of 1 and 2 are estimated to be 1.37 and 1.46 eV, respectively (Figure 4d, Supporting Information). These bandgap energies are significantly lower than those of porphyrinic MOFs such as PCN-222 (1.75 eV) and PCN-601 (1.75 eV).^[20,31,52,71,72] Mott-Schottky measurements at a frequency of 1000 Hz were used to characterize the semiconducting properties of MOFs and their potential for CO₂ photoreduction. As exhibited in Figure 4e,f, the positive slope of the curves indicates that both MOFs 1 and 2 exhibit n-type semiconductor properties.^[31,59,71] The *E*_{fb} of MOF 1 is −0.37 V versus an Ag/AgCl electrode (−0.37 V vs NHE, pH 7). The *E*_{fb} of n-type semiconductors is considered to be 0.1–0.2 V lower than the CBM, so the CBM of MOF 1 is estimated to be −0.47 to −0.57 V versus NHE. Taking the bandgap value into account, the VBM of MOF 1 is calculated to be +0.80 to +0.90 V versus NHE. For MOF 2, the CBM is evaluated to be −0.50 to −0.60 V versus NHE, and the VBM is calculated to be +0.86 to +0.96 V versus NHE. Clearly, the redox potentials of CBM of MOFs 1 and 2 are more negative than that of CO/CO₂ (−0.48 V vs NHE, pH 7) and CH₄/CO₂ (−0.24 V vs NHE, pH 7).^[20,26,37,71,73] Therefore,

the band alignments of MOFs 1 and 2 mean that photocatalytic CO₂ reduction is thermodynamically feasible.

2.5. Catalytic Performance for CO₂ Photoreduction

Considering the high porosity, wide-ranging light harvesting properties, catalytically active metal composition, and appropriate band alignment of MOFs 1 and 2, it is speculated that they have good application prospects for photocatalytic CO₂ reduction. Both the MOFs 1 and 2 catalysts exhibited significant photocatalytic CO₂ reduction activity without any cocatalyst or photosensitizer. As shown in Figure 5a, almost no product formation was detected in the dark environment during the first hour. Under the irradiation of a xenon lamp, the amount of generated CO increased rapidly and almost linearly with increasing irradiation time. The monolayer MOF 1 exhibited a high CO release rate, and no other gaseous or liquid by-products except H₂ and a very small amount of CH₄ were detected in this catalytic system. The evolution amounts of CO, H₂, and CH₄ in 6 h increased to 365.59, 52.13, and 3.14 μmol g⁻¹, respectively. The selectivity of CO over H₂ and CH₄ formation was as high as 86.87%. Impressively, the only product of the CO₂ reduction reaction on bilayer MOF 2 was CO; neither CH₄ nor H₂ was detected (Figure 5b). The average CO yield was 276.89 μmol g⁻¹ under light for 6 h. This CO yield is higher than those of other porphyrin-based MOFs (such as PCN-601, PCN-222, and Ni₃TCPP) and classical inorganic Pt/CdS photocatalysts (Figure 5f).^[20,31] Furthermore, the 100% CO selectivity of bilayer MOF 2, which is higher than that of most reported

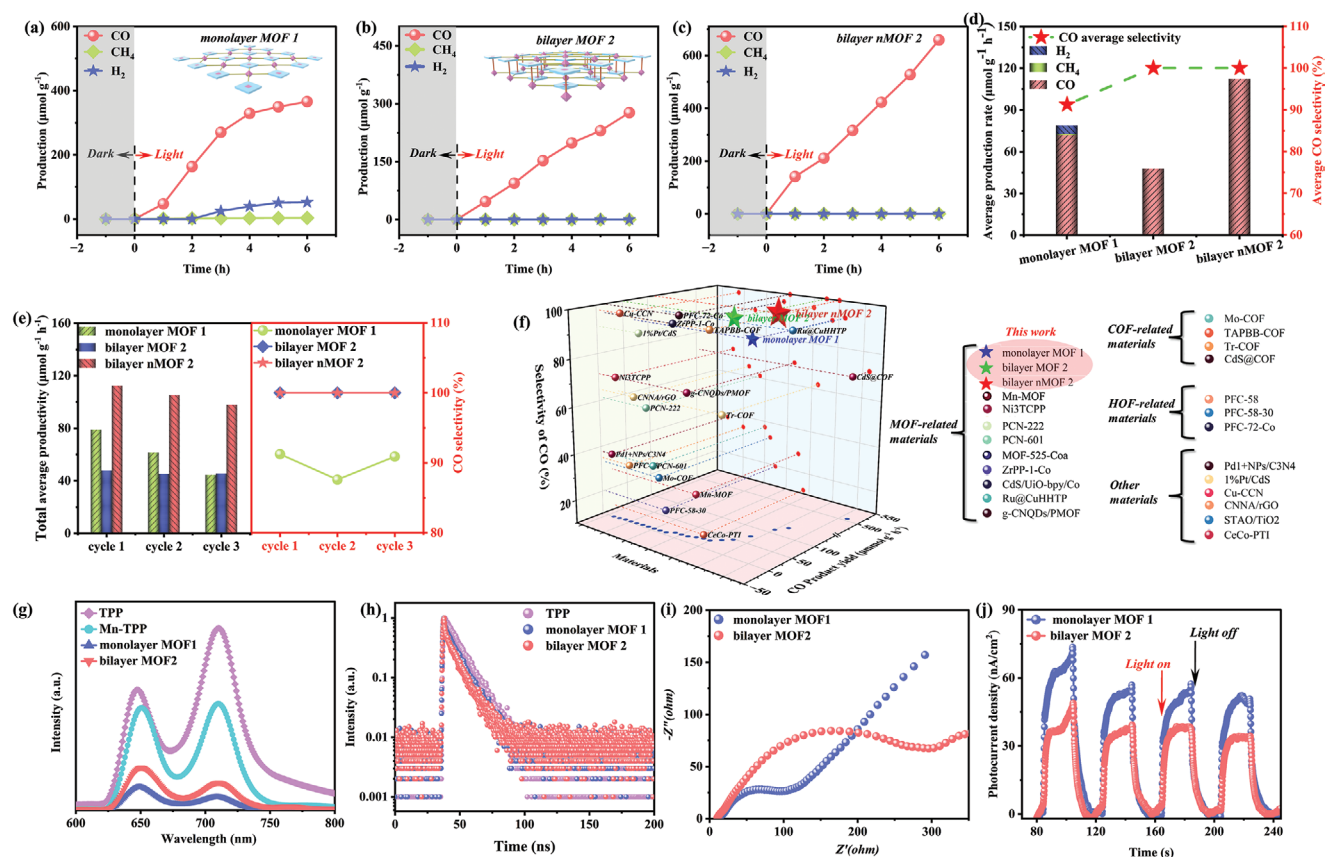


Figure 5. Photocatalytic performances of MOFs for CO₂ reduction: a–c) Time-dependent CO, CH₄, and H₂ evolution upon visible-light irradiation for monolayer MOF 1, MOF 2, and nMOF 2, respectively, d) production rates, e) total yield and selectivity of three reaction cycles in 6 h, f) comparison with the catalysis performance of reported different photocatalysts (without cocatalyst or photosensitizer), g) PL emission excitation, h) time-resolved PL decay of MOFs 1 and 2 under excitation at 710 nm, i) electrochemical impedance spectra (EIS) of MOFs 1 and 2, j) transient photocurrents of MOFs 1 and 2.

MOFs materials or other classes of materials^[20,31,41,56,58,71,59,74–86] (Figure 5f and Table S1, Supporting Information), means that subsequent product separation is not required, significantly reducing the potential cost of operation. Next, control experiments were used to verify the excellent photocatalytic CO₂ reduction of the MOFs. Products were barely detected in the absence of the MOF catalysts or TEOA, suggesting that both the photocatalysts and electron donors are essential for CO₂ reduction (Figures S23 and S24, Supporting Information). In addition, when Ar was used instead of CO₂, no CO products were detected in the photoreduction system of MOFs 1 and 2, and only a small amount of H₂ was detected in the MOF 1. The negligible CO release confirms that CO originates from CO₂ molecules.

Recyclability is one of the important indicators for evaluating the quality of catalysts. To confirm whether MOFs 1 and 2 materials can maintain high activity across multiple cycles, the recovered catalysts were added to the photocatalytic reaction system again, and the experiment was repeated. As shown in Figure 5e, MOF 2 exhibited high durability in this cycling test, with negligible loss of catalytic activity after three consecutive cycles. In the third cycle experiment, MOF 1 still maintained a high catalytic activity, but catalytic activity decreased to a certain extent. We speculate that this decline in the catalytic activity of

MOF 1 should be attributed to its poor structural stability and some of the MOF degrading during the catalytic reaction. The cycling stability of these materials was further verified by examining the recovered catalysts. The XRD pattern of the recovered MOF 2 confirmed that its crystal phase structure remained basically unchanged after the catalytic reaction (Figure 3h). Furthermore, IR (Figures S11 and S12, Supporting Information) and XPS (Figures S26 and S27, Supporting Information) further confirmed the stability of MOF 2. Given the good cycling stability of MOF 2, we further explored its chemical stability, as this is important for the use of materials in harsh conditions. Therefore, the acid-base stability of MOF 2 was evaluated by treating the samples under different conditions (pH = 1–10, soaking for 24 h). Notably, the XRD patterns of treated MOF 2 were observed to be unchanged, indicating no phase transition or structural collapse during treatment (Figure 3h). This high acid-base stability means that MOF 2 is more promising as a catalyst for practical photocatalytic CO₂ reduction. Therefore, compared with the monolayer MOF 1, the bilayer MOF 2 exhibits higher selectivity and better durability, and this MOF shows good promise for application in large-scale photocatalytic CO₂ reduction.

Furthermore, considering that MOF nanosheets (nMOFs) have higher surface areas and more exposed surface sites than

their bulk counterparts, the use of nMOFs can facilitate contact between surface sites and substrate molecules, leading to enhanced catalytic performance. nMOF 2 has a higher BET specific surface area of 346.99 m² g⁻¹ relative to MOF 2, and the N₂ adsorption-desorption isotherm is shown in Figure S28, Supporting Information. In addition, the CO₂ adsorption isotherms collected at 273 and 298 K (1 bar) showed maximum adsorption of 51.64 and 31.78 cm³ g⁻¹, the *Q*_{st} of CO₂ was 36.47 kJ mol⁻¹ (Figure S29, Supporting Information), both significantly higher than that of MOF 2. Consistent with our conjecture, nMOF 2 exhibited higher photocatalytic CO₂ reduction efficiency compared to the bulk MOF 2, with a CO generation rate of 658.74 μmol g⁻¹ within 6 h (Figure 5c). This CO production rate is 2.36 times higher than that of the bulk MOF 2. We propose that if MOF nanosheets with more uniform thickness and thinner thickness can be obtained, the efficiency of CO₂ photoreduction will be further improved.

2.6. Mechanism of Excellent Photocatalytic CO₂ reduction performance

From the perspective of charge dynamics, the photocatalytic process mainly includes charge generation, charge transfer, and charge consumption.^[87] Catalysts that absorb large-scale solar energy can optimize charge generation by increasing the number of electrons transformed to photogenerated. If electron-hole recombination can be suppressed, more electrons and holes can reach the catalyst surface and participate in redox reactions. Moreover, efficient charge depletion helps inject more charges into the reduction reaction. Therefore, the adsorption and activation of CO₂ on the catalyst surface is the key to charge consumption. Next, we will discuss in detail the intrinsic mechanism of the excellent photocatalytic CO₂ reduction performance of MOFs 1 and 2.

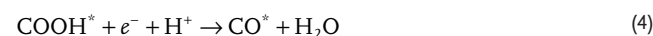
2.6.1. Charge Transfer and Separation Kinetics

The photogenerated charge transfer kinetics of the MOFs 1 and 2 for CO₂ reduction were investigated through photo/electrochemical characterization. As shown in Figure 5g, the PL intensities of MOFs 1 and 2 are significantly quenched relative to the intensities of the TPP ligand, with values only 7.91% and 14.8% of that of TPP, respectively (*λ*_{ex} = 520 nm). It is well known that a higher PL intensity means a greater possibility of charge carrier recombination. This significant quenching of PL intensity indicates that radiative electron-hole recombination in these MOFs is significantly suppressed and charge separation is effectively improved. To investigate charge separation in more detail, time-resolved PL (TRPL) spectra of the MOFs were obtained. Figure 5h shows the normalized TRPL plots. Upon excitation at 375 nm, the average PL lifetime at 710 nm for MOF 1 increases from 9.39 to 10.54 ns compared to MOF 2, indicating that the recombination of photogenerated charges is suppressed. Thus, the carrier lifetime is enhanced. Furthermore, electrochemical impedance spectra (EIS) were obtained to reflect the charge-transfer kinetics at the electrode interface. As shown in Figure 5i, the EIS curves of MOFs 1 and 2 both

exhibit semicircles in the high-frequency region. These small semicircles correspond to the charge transfer resistance. The semicircular radius of the high-frequency region of MOF 1 is smaller than that of MOF 2, indicating that MOF 1 has lower charge transfer resistance. This demonstrates that MOF 1 has a high electron-hole pair separation speed and interfacial charge transfer speed. To further elucidate the charge generation and separation efficiencies, the transient photocurrent responses of MOFs 1 and 2 were measured under visible light irradiation. As shown in Figure 5j, MOF 1 has a higher photocurrent response compared to MOF 2, indicating better charge generation and separation efficiency, which is consistent with the trend of their catalytic efficiency. These results show that effectively improving the charge transport capacity of photocatalysts is beneficial for improving the overall efficiency of the photocatalytic CO₂ reduction reaction.

2.6.2. CO₂ Reduction Pathways and Mechanisms

In general, the photocatalytic CO₂ reduction reaction can be described with the following four processes: 1) CO₂ molecules are adsorbed on the photocatalyst surface to form CO₂*; 2) the protons generated by water splitting interact with the CO₂* adsorbed on the surface to form a COOH* intermediate; 3) through a proton-electron transfer process, COOH* is reduced to CO*; 4) the CO* on the catalyst surface is desorbed to generate CO.



"*" represents the active adsorption state.

First, the adsorption energies of CO₂ and H₂O on the MOF materials were investigated by DFT calculations (Table 2). For both the monolayer MOF and bilayer MOF, the initial adsorption of CO₂ is exothermic, and CO₂ molecules are more easily adsorbed on MOF materials. The adsorption energy of CO₂ on the monolayer MOF is -0.702 eV, and the adsorbed CO₂ molecules adopt a curved configuration to bind the Mn metal ion

Table 2. The adsorption energies (*E*_{ads}, eV) of intermediates to the central Mn in porphyrin along the optimal reaction path on the monolayer MOF 1 and bilayer MOF 2.

	Monolayer MOF 1	Bilayer MOF 2
<i>E</i> _{ads} (*CO ₂)	-0.70 eV	-0.31 eV
<i>E</i> _{ads} (*COOH)	-0.93 eV	-0.91 eV
<i>E</i> _{ads} (*CO)	-2.77 eV	-1.81 eV
<i>E</i> _{ads} (*H ₂ O)	-0.99 eV	-0.62 eV

"*" represents the active adsorption state.

sites in the porphyrin center. This indicates a strong interaction between the Mn center and CO₂. In comparison, the adsorption energy of CO₂ on the bilayer MOF is −0.313 eV, indicating that the interaction between the Mn metal center and CO₂ is weaker for this bilayer MOF than for the monolayer MOF. This is further demonstrated by the linear geometry of the adsorbed CO₂ on the bilayer MOF. In addition, the adsorption energy of H₂O on the monolayer MOF (−0.989 eV) is significantly stronger than that of H₂O on the bilayer MOF (−0.622 eV). It is generally believed that stronger CO₂ adsorption energy is beneficial for stabilizing the initial metal CO₂ adduct, which promotes the reduction of CO₂. A weaker H₂O adsorption energy can suppress unwanted metal H₂O adducts, thereby reducing H₂ by-products. This is the same order as the CO yield and selectivity in the experimental results. To more thoroughly investigate the CO₂ reduction process on these MOFs, the total potentials for CO₂ reduction to CO via COOH* on both the monolayer MOF and bilayer MOF were further calculated by DFT calculations. The potential energy barriers for the reduction of CO₂*

to COOH* are −0.359 and −0.018 eV for the monolayer MOF and bilayer MOF, respectively. Both of these values are small enough to indicate that the reaction can proceed at room temperature. Figure 6a shows that the reduction of COOH* to the CO* adduct easily occurs in both catalytic systems. DFT calculations for the monolayer MOF indicate that the activation energy barrier for CO* formation is −1.885 eV, which is lower than that of the bilayer MOF (−0.908 eV). This suggests that the monolayer MOF is kinetically more favorable than the bilayer MOF for CO* formation, leading to the enhanced photocatalytic performance of the monolayer MOF. Moreover, this is in the same order as the CO production rate. The strong adsorption affinity of MOFs 1 and 2 for CO₂ favors the stabilization of the initial Mn-CO₂ adduct, thereby promoting the conversion of CO₂ to CO. The weak adsorption affinity of MOF 2 for H₂O suppresses proton transfer and reduces the formation of H₂ by-products. Therefore, stronger CO₂ affinity and weaker H₂O affinity are expected to jointly improve the photoreduction activity and selectivity of MOF photocatalysts.

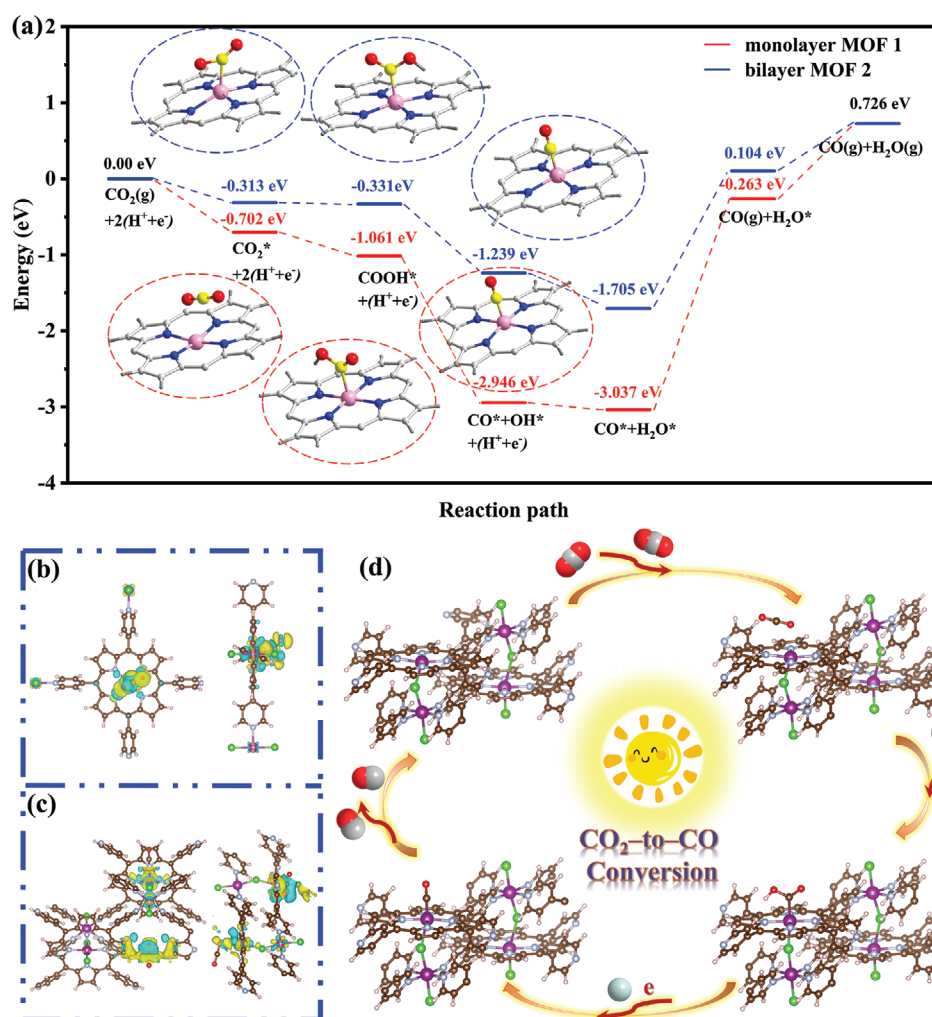


Figure 6. a) Computational free energy diagram of CO₂ reduction reaction for monolayer MOF 1 and bilayer MOF 2. b,c) Charge densities of CO₂* intermediates over monolayer MOF 1 and bilayer MOF 2. The light yellow and lake blue contours represent charge depletion and charge accumulation, respectively. The color-based atomic labels are C: brown; N: blue; H: white; Mn: purple; Cl: green; O: red. d) DFT simulation process of the conversion of CO₂ to CO (taking bilayer MOF 2 as an example).

Based on the experimental characterization and theoretical calculations, a photo-driven catalytic mechanism for the reduction of CO₂ on the monolayer and bilayer MOFs is rationally proposed. In short, in the photocatalytic CO₂ reduction process, the metal Mn atom in the porphyrin center acts as an affinity site for enhanced CO₂ adsorption and an effective catalytic active site, which effectively improves the CO₂ reduction activity of the MOF materials. The moderate CO₂ adsorption energy of MOFs 1 and 2 suggest that these materials possess the desired CO₂ affinity and can be easily regenerated with low energy loss, which gives them a kinetic advantage for promoting CO₂ photocatalysis. Based on the analysis reported herein, the porphyrin ligand acts as a visible light harvesting unit, generating electron-hole pairs upon irradiation and then rapidly transferring photogenerated electrons to the porphyrin metal center. The CO₂ adsorbed on the active site is reduced to CO by energetic electrons. The holes are trapped by TEOA, completing the catalytic cycle (Figure 6d).

3. Conclusion

In summary, we successfully constructed a monolayer MOF 1 and a novel dislocated 2D bilayer MOF 2 under simple solvothermal conditions by adjusting the temperature of the reaction system and the polarity of the solvent. Due to their excellently visible light harvesting ability, atomically dispersed metal active sites, and sufficient redox potential, these MOFs can serve as efficient photocatalysts for the CO₂ reduction reaction. The structural difference between the monolayer MOF 1 and bilayer MOF 2 lead to their distinct electronic and optical properties, which in turn lead to variance in photocatalytic CO₂ reduction performance. Both the monolayer MOF and bilayer MOF significantly promote the photoreduction of CO₂ to CO without the use of any cocatalyst or photosensitizer, with CO yields of 365.59 and 276.89 $\mu\text{mol g}^{-1}$, respectively. It is worth noting that the bilayer MOF 2 exhibits ultra-high CO selectivity and can convert CO₂ to CO with 100% selectivity, which effectively avoided secondary separation after photocatalytic reaction. Moreover, the high cycling stability and chemical stability of the bilayer MOF 2 also demonstrate its good promise for large-scale photocatalytic CO₂ reduction. Furthermore, MOF 2 nanosheets also exhibit higher catalytic activity compared to its bulk counterpart, due to their more exposed surface sites. DFT calculations show that increasing the adsorption energy of the photocatalyst for CO₂ and decreasing the adsorption energy for H₂O are effective strategies for improving catalytic activity and selectivity for CO₂ reduction. This work proposes a new strategy for the construction of 2D bilayer MOFs, illuminating a new pathway for the controllable synthesis of novel 2D MOFs, and also provides a new perspective for the customization of photocatalytic performance.

4. Experimental Section

Synthesis of Monolayer MOF 1: TPP (0.004 mmol) and MnCl₂ (0.020 mmol) were added to a mixed solvent of 4 mL NMP and EtOH (v/v = 3/1). This mixture was transferred into a 25 mL Teflon-lined

stainless-steel vessel and reacted at 160 °C for 3 days. After gradually cooling to room temperature, dark green flaky Mn-MOF 1 crystals were obtained by filtration. These crystals were washed with ethanol and naturally air-dried. The yield was 10% (based on TPP).

Synthesis of Bilayer MOF 2: The synthetic procedure of MOF 2 was the same as that of 1 except that NMP/EtOH (v/v = 3/1) was replaced by DMF/EtOH (v/v = 3/1) and the reaction temperature was lowered from 160 to 110 °C. Dark green square crystals of 2 were acquired with a yield of 48% (based on TPP).

Synthesis of MOF 1 and 2 nanosheets: Crystals of the MOF 1 or 2 crystals were ground to ensure homogeneity. Next, 2 mg crystal powder was added to a 20 mL glass vial along with 10 mL absolute ethanol. Uniformly dispersed MOFs nanosheets were obtained after sonicating the samples with an ultrasonic cleaner (UA06MFD, ultrasonic power: 260 W) at a frequency of 40 kHz for 2 h. The temperature was maintained at ≈ 25 °C during sonication.

Electrochemical Measurements: For preparation of working electrode to carry out the electrochemical characterization, the indium tin oxide (ITO) glass (1 $\times$ 2 cm²) was used as the supporting substrate and is first washed by ethanol and water under ultrasonic processing for about 30 min, and then dried by sweeping gas of N₂. The sample suspension was prepared by dispersing 2 mg of catalyst in the solution containing 1 mL of ethanol and 10 μL Nafion followed by ultrasonication for 60 min. The working electrode was prepared by dropping the sample suspension (200 μL) onto the surface of an ITO plate and drying it naturally (Coated area of 1 $\times$ 1 cm²). All electrochemical tests were measured using an electrochemical analyzer (CHI760E) on a standard three-electrode cell with photocatalyst-coated ITO as the working electrode, Pt wire as the counter electrode, and Ag/AgCl as the reference electrode. The Mott-Schottky tests used Na₂SO₄ solution (0.1 M) and the EIS tests used a mixed aqueous solution of 5 mM K₃[Fe(CN)₆]/5 mM K₄[Fe(CN)₆]/0.1 M KCl.

Photoelectrochemical Measurements: The photocatalytic reaction was carried out in a mixed solution of MeCN, H₂O and TEOA (10 mL, MeCN/H₂O/TEOA = 8:2:1) containing MOFs or MOF nanosheets (5 mg) as the photocatalyst for CO₂ reduction, with TEOA as the electron donor. The reaction was carried out in a custom-made 20 mL photocatalytic reaction cell. The photocatalytic reaction system was a solid-liquid mode heterogeneous reaction system, and the reaction process was a flow experiment process. Degassed and saturated with ultrapure CO₂ to remove any dissolved O₂, and then the reaction cell was sealed. This reactor is connected with a PerfectLight Labsolar-IIIAG system, which was used to circulate the feed gas of CO₂ and the gas of products, and at the same time was used as online gas sampler to quantitatively analyze the gas product by gas-chromatography. After the reaction apparatus was pre-equilibrated in the dark for 1 h, the reaction was carried out under 300 W Xe arc lamp (PerfectLight, PLS-SXE300C), and the light intensity was adjusted at 1 sun (AM1.5) using a 1 sun checker. A cooling circulating water system was used to stabilize the reaction temperature at 25 °C. Photocatalytic reduction products were recorded per hour for 6 h using GC9700 equipped with TCD detectors. The used catalyst was dried at 80 °C for 6 h to activate before being used in the next examination. The recycle test was examined in the completely identical environment.

The control experiment was carried out using high pure Ar instead of CO₂ as the reaction atmosphere in the identical condition. No signals corresponding to CO or CH₄ can be detected during the 2 h control reactions. It confirmed the CO₂ related product was from CO₂ and excluded the possibility from decomposed catalyst, reactant reagent or the reaction system.

The CO selectivity is calculated as Equation 6:

$$\text{CO selectivity} = \frac{n_{\text{CO}}}{n_{\text{CO}} + n_{\text{H}_2}} \quad (6)$$

Sample Activation and Gas Adsorption: The specific surface area and pore properties of the MOFs were investigated by N₂ adsorption experiments. Before the experiment, freshly prepared samples were soaked in methanol for exchange for 3 days and vacuum-dried at 120 °C for 5 h to remove the lattice solvent.

CCDC 2 189 222 and 2 189 221 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

2D MOFs, bilayer, high selectivity, photocatalytic CO₂ reduction

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